SRI SHAKTHI SILICON INITIATIVE

SRI SHAKTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

(AN AUTONOMOUS INSTITUTION)

Approved by AICTE, New Delhi · Affiliated to Anna University, Chennai · TNEA Counselling Code: 2727

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DEPARTMENT OF EE - VDT

ELECTRONICS ENGINEERING (VLSI DESIGN AND TECHNOLOGY)

Supported by MeitY · Chips to Startup (C2S) Programme · Government of India

NATIONAL-LEVEL 2-STAGE TECH & SILICON HACKATHON

VELTRAXX 2.0

BUILD · SYNTHESIZE · DEPLOY · 24 HOURS OF RAW SILICON INNOVATION

OFFICIAL TECHNICAL BENCHMARK SPECIFICATION

OFFICIAL PROBLEM STATEMENTS COMPENDIUM

Comprehensive Smart India Hackathon (SIH) Benchmark Edition spanning 25 Industrial Hardware, ASIC, FPGA, Edge AI, and Cyber-Physical Systems Challenges.

EVENT STAGE	OFFICIAL SCHEDULE & WINDOW	MODE & PARTICIPATION GUIDELINES
STAGE 01 **Round 1 Registration Close**	**01 December 2026 (11:59 PM IST)**	**₹0 Free Entry** · Online squad registration across 5 tracks
STAGE 02 **48-Hour Solution Blueprint Upload**	**02 Dec (00:00) – 03 Dec 2026 (23:59 IST)**	Online 6-Slide Technical Presentation PDF Submission
STAGE 03 **24-Hour Offline Grand Finale**	**15 – 16 December 2026 (24H Non-Stop)**	SIET Tech Park Arena, Coimbatore · Cash Prize Defense

COMPREHENSIVE DIRECTORY

MASTER INDEX OF PROBLEM STATEMENTS

All 25 problem statements adhere to the Smart India Hackathon (SIH) rigorous evaluation benchmark. Each problem statement occupies a dedicated page from Page 3 to Page 27.

TIER A (JUNIOR DIVISION · 1ST & 2ND YEAR UG)

Graded on architectural logic, functional block diagrams, truth tables, and behavioral simulation (Icarus Verilog, EDA Playground, Proteus, Wokwi, Python).

TIER B (SENIOR DIVISION · 3RD & 4TH YEAR UG, PG & SCHOLARS)

Graded on synthesizable RTL, synthesis PPA reports (Cadence Genus / Synopsys / Vivado / Yosys), timing/power closure, FPGA hardware, and fault injection.

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VTX-T1-03	Track 01	**Fault-Tolerant 5-Port Network-on-Chip (NoC) Mesh Router**	Hardware / Multi-Core Interconnects	P.5
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TRACK 01: CORE ASIC & CUSTOM SILICON ARCHITECTURE (VLSI-A)

VTX-T1-01 Low-Power Configurable ALU with Autonomous Logic BIST (LBIST)

DOMAIN TRACK	**Track 01: Core ASIC & Custom Silicon Architecture (VLSI-A)**
ORGANIZING HOST	**Department of EE - VDT, Sri Shakthi Institute of Engineering and Technology**
CATEGORY	**Hardware / RTL Design, Design-for-Testability (DFT) & Low-Power VLSI**

1. BACKGROUND & INDUSTRY CONTEXT

REAL-WORLD CHALLENGE

In automotive, aerospace, and mission-critical System-on-Chip (SoC) architectures (such as ISO 26262 ASIL-D compliant controllers), arithmetic execution units cannot rely solely on factory Automated Test Equipment (ATE) to detect silicon defects. Aging effects, electromigration, and latent manufacturing defects require the chip to test itself autonomously during every power-on cycle and idle window using **Logic Built-In Self-Test (LBIST)**. However, traditional single-MISR (Multiple Input Signature Register) architectures suffer from *fault aliasing* (where two faulty bit sequences cancel each other out to produce a valid golden signature) and excessive dynamic switching power during normal idle states.

2. PROBLEM DESCRIPTION & SCOPE OF INNOVATION

TECHNICAL OBJECTIVE

Teams must architect and verify a synthesizable, multi-operation **Arithmetic Logic Unit (ALU)** integrated with an on-chip **Autonomous LBIST Controller**. When `bist_mode = 1`, the system must isolate the ALU from external system buses, automatically feed pseudo-random test vectors generated by a Linear Feedback Shift Register (LFSR), capture the resulting ALU outputs plus status flags (`Zero`, `Carry`, `Overflow`, `Negative`) into a signature compaction engine, and compare the final signature against a hardcoded golden reference to assert `fault_detected` or `test_pass`. During normal functional mode (`bist_mode = 0`), unused arithmetic sub-blocks must be dynamically isolated or clock-gated to minimize dynamic power dissipation.

3. EXPECTED SYSTEM ARCHITECTURE & FUNCTIONAL SPECIFICATIONS

DELIVERABLES

1. **Core ALU Datapath:** Must support at least 8 distinct arithmetic, logical, and barrel-shift operations with registered inputs/outputs and status flag generation.
2. **LFSR Pattern Generator:** Maximal-length primitive polynomial LFSR generating deterministic pseudo-random operand pairs (`A`, `B`, and `Opcode`) across a configurable number of test clock cycles (e.g., 256 or 1024 cycles).
3. **BIST FSM Controller:** Autonomous state machine (`IDLE -> ISOLATE -> PATTERN_RUN -> COMPARE_SIGNATURE -> REPORT_DONE`) controlled via a single `start_bist` trigger pin.
4. **Fault-Aliasing Reduction & Power Optimization:** Separation of odd/even cycle signatures (Dual-MISR) or bit-swapping LFSR to reduce switching activity and prevent signature aliasing.

TIER A: JUNIOR DIVISION (1ST & 2ND YEAR UG)

Synthesizable 8-bit or 16-bit Verilog/SystemVerilog RTL implementation of the ALU, LFSR, BIST FSM, and single MISR.

Self-checking testbench demonstrating both normal ALU mode and a simulated stuck-at-0/stuck-at-1 fault injection inside the ALU that triggers `fault_detected = 1`.

TIER B: SENIOR DIVISION (3RD & 4TH YEAR UG / PG)

32-bit pipelined ALU with Dual-MISR (odd/even cycle compaction) and operand isolation / integrated clock-gating (ICG) for low dynamic power.

Synthesis PPA (Power, Performance, Area) reports (Cadence Genus / Synopsys Design Compiler / Vivado / Yosys) and Formal Equivalence or fault-coverage analysis.

SLIDE 5 **MANDATORY PROOF-OF-COMPETENCE ARTIFACT (FOR ROUND 1 PPT)**

Your Slide 5 MUST include: **(a)** The exact primitive polynomial equation used for your LFSR/MISR (e.g., `P(x) = x^{16} + x^{14} + x^{13} + x^{11} + 1`), **(b)** Your BIST Controller FSM state diagram, and **(c)** A preliminary simulation waveform or truth table showing signature divergence under a single stuck-at fault.

TRACK 01: CORE ASIC & CUSTOM SILICON ARCHITECTURE (VLSI-A)

VTX-T1-02 Metastability-Hardened Dual-Clock Asynchronous FIFO Bridge

DOMAIN TRACK	**Track 01: Core ASIC & Custom Silicon Architecture (VLSI-A)**
ORGANIZING HOST	**Department of EE - VDT, Sri Shakthi Institute of Engineering and Technology**
CATEGORY	**Hardware / Clock Domain Crossing (CDC) & SoC Interconnects**

1. BACKGROUND & INDUSTRY CONTEXT — REAL-WORLD CHALLENGE

Modern heterogeneous SoCs contain dozens of independent clock domains—for example, a high-speed 800 MHz processor core communicating with a 50 MHz SPI/UART peripheral bus or a 125 MHz Ethernet MAC. Passing multi-bit counters or data words directly across unrelated clock boundaries causes setup/hold timing violations, resulting in **metastability** where flip-flops oscillate in an indeterminate voltage state and corrupt downstream logic.

2. PROBLEM DESCRIPTION & SCOPE OF INNOVATION — TECHNICAL OBJECTIVE

Teams must design and verify a parameterized, **Metastability-Hardened Dual-Clock Asynchronous FIFO Bridge** capable of streaming burst data safely between an asynchronous Write Clock domain (`wclk`, `wrst_n`) and an independent Read Clock domain (`rclk`, `rrst_n`). The architecture must guarantee zero data loss, zero phantom reads/writes, and glitch-free synchronization of read and write pointers across any frequency ratio (fast-to-slow, slow-to-fast, or irrational clock frequencies).

3. EXPECTED SYSTEM ARCHITECTURE & FUNCTIONAL SPECIFICATIONS — DELIVERABLES

1. **Dual-Port Memory Array:** Parameterized data width (`DATA_WIDTH`, default 16 or 32 bits) and depth (`ADDR_WIDTH`, `2^N` locations).
2. **Binary-to-Gray & Gray-to-Binary Pointer Logic:** Because standard binary counters change multiple bits simultaneously (e.g., `0111` to `1000` flips 4 bits at once), read and write pointers must be converted into **Gray Code** (where only 1 bit changes per increment) before crossing clock boundaries.
3. **Multi-Stage Flip-Flop Synchronizers:** 2-stage (or configurable 3-stage) D-flip-flop synchronizer chains transferring Gray-coded pointers across `wclk` and `rclk`.
4. **Full, Empty & Programmable Watermark Generation:** Accurate generation of `wfull`, `rempty`, `almost_full`, and `almost_empty` flags using MSB/2nd-MSB Gray pointer comparison logic.

TIER A: JUNIOR DIVISION (1ST & 2ND YEAR UG)

Synthesizable Verilog RTL for an 8-bit or 16-bit deep Asynchronous FIFO with Gray-code conversion and 2-stage synchronizers.

Dual-clock testbench demonstrating burst write until `wfull = 1` and burst read until `rempty = 1` across two unrelated clock periods (e.g., `T_{wclk} = 7\text{ ns}` and `T_{rclk} = 13\text{ ns}`).

TIER B: SENIOR DIVISION (3RD & 4TH YEAR UG / PG)

Parameterized RTL supporting non-power-of-two depths or dynamic programmable `almost_full` / `almost_empty` thresholds, plus asynchronous reset assertion / synchronous de-assertion circuits (`reset_sync`).

Static Timing Analysis (STA) constraints (`set_false_path` / `set_max_delay` on CDC paths) and synthesis reports proving MTBF (Mean Time Between Failures) resilience.

SLIDE 5 MANDATORY PROOF-OF-COMPETENCE ARTIFACT (FOR ROUND 1 PPT)

Your Slide 5 MUST include: **(a)** A complete block diagram showing the exact `wclk` and `rclk` domain boundary line, **(b)** The exact boolean/XOR equation used to detect the **Full Condition** in Gray code (comparing the top two MSBs vs. remaining bits), and **(c)** Preliminary dual-clock simulation waveforms.

TRACK 01: CORE ASIC & CUSTOM SILICON ARCHITECTURE (VLSI-A)

VTX-T1-03 Fault-Tolerant 5-Port Network-on-Chip (NoC) Mesh Router

DOMAIN TRACK	**Track 01: Core ASIC & Custom Silicon Architecture (VLSI-A)**
ORGANIZING HOST	**Department of EE - VDT, Sri Shakthi Institute of Engineering and Technology**
CATEGORY	**Hardware / Multi-Core Interconnects, NoC & Fault-Tolerant Silicon**

1. BACKGROUND & INDUSTRY CONTEXT

REAL-WORLD CHALLENGE

As AI accelerators and multi-core processors scale beyond 8 to 64 cores on a single die, traditional shared buses (like AXI crossbars) become wiring bottlenecks due to capacitive loading and arbitration stalls. Industry chips use packet-switched **Network-on-Chip (NoC)** mesh routers. However, at deep sub-micron nodes, crosstalk and cosmic-ray single-event upsets (SEUs) flip bits inside router buffers, causing **Silent Data Corruption (SDC)** or deadlocked packets routed to the wrong core.

2. PROBLEM DESCRIPTION & SCOPE OF INNOVATION

TECHNICAL OBJECTIVE

Teams must design a synthesizable **5-Port Network-on-Chip (NoC) Mesh Router** (`North`, `South`, `East`, `West`, and `Local PE Core` ports) that routes flits/packets across a 2D mesh coordinate grid `(X, Y)` without deadlock, while actively detecting and recovering from bit-flip corruptions inside the packet header and payload.

3. EXPECTED SYSTEM ARCHITECTURE & FUNCTIONAL SPECIFICATIONS

DELIVERABLES

1. **Packet/Flit Structure:** Defined packet format containing `Header` (Source X/Y, Destination X/Y, Packet ID) + `Payload` + `Error Check Code (ECC / Parity)`.
2. **Deadlock-Free XY Routing Engine:** Compares current router coordinates `(X_curr, Y_curr)` with target coordinates `(X_dest, Y_dest)`—routing horizontally (`East` / `West`) first until `X_curr == X_dest`, then vertically (`North` / `South`), and finally ejecting at `Local`.
3. **Round-Robin / Priority Switch Allocator:** Fair arbitration when two input ports simultaneously request the same output port (preventing port starvation).
4. **Silent Data Corruption (SDC) Guard:** Hardware parity or Hamming SEC-DED (Single Error Correction, Double Error Detection) encoder/decoder on router links with automatic single-bit correction or `NACK` retransmission request.

TIER A: JUNIOR DIVISION (1ST & 2ND YEAR UG)

Synthesizable RTL for a 5-port router with input FIFO buffers, deterministic XY routing engine, 5x5 crossbar switch, and even/odd parity error detection flag.

Simulation testbench sending packets simultaneously across multiple ports to different `(X, Y)` coordinates.

TIER B: SENIOR DIVISION (3RD & 4TH YEAR UG / PG)

Wormhole switching with Virtual Channel (VC) flow control (`credit/valid` handshaking), Round-Robin arbiter, and Hamming SEC-DED single-bit hardware auto-correction on every flit.

Synthesis area/timing breakdown and latency-vs-injection-rate throughput analysis.

SLIDE 5 **MANDATORY PROOF-OF-COMPETENCE ARTIFACT (FOR ROUND 1 PPT)**

Your Slide 5 MUST include: **(a)** The exact bit-level packet/flit frame layout (e.g., 32-bit flit field bit-positions), **(b)** The XY routing decision truth table, and **(c)** The arbiter priority state diagram or Hamming parity matrix.

TRACK 01: CORE ASIC & CUSTOM SILICON ARCHITECTURE (VLSI-A)

VTX-T1-04 Dynamic Clock-Gated Multi-Master APB/AXI-Lite Peripheral Interconnect

DOMAIN TRACK	**Track 01: Core ASIC & Custom Silicon Architecture (VLSI-A)**
ORGANIZING HOST	**Department of EE - VDT, Sri Shakthi Institute of Engineering and Technology**
CATEGORY	**Hardware / AMBA SoC Protocols & Low-Power Clock Gating**

1. BACKGROUND & INDUSTRY CONTEXT — REAL-WORLD CHALLENGE

In battery-powered IoT SoCs and wearables, the clock tree accounts for 35% to 50% of total dynamic power dissipation (`P_{dyn} = \alpha \cdot C \cdot V_{DD}^2 \cdot f`). Even when low-speed peripherals (UART, SPI, I2C, PWM, Watchdog Timer) are not actively being addressed by the CPU, their internal registers continue toggling on every clock edge unless an intelligent **Peripheral Bus Interconnect** dynamically gates their clocks and isolates their data buses.

2. PROBLEM DESCRIPTION & SCOPE OF INNOVATION — TECHNICAL OBJECTIVE

Teams must architect an **AMBA-Compliant Peripheral Interconnect & Power-Aware Bridge** that connects one or more bus masters to at least 4 memory-mapped peripheral slaves, automatically enabling the clock (`PCLK_gated[i]`) ONLY for the specific peripheral slave addressed during an active bus transaction, while keeping all idle slaves in a zero-toggle clock-gated sleep state. Furthermore, the bridge must include a hardware **Bus Watchdog Timeout Counter** so a hung peripheral (`PREADY` stuck at `0`) never freezes the entire processor bus.

3. EXPECTED SYSTEM ARCHITECTURE & FUNCTIONAL SPECIFICATIONS — DELIVERABLES

1. **Address Decoder & Memory Map:** Clean address decoding mapping distinct base address ranges (e.g., `0x4000_0000` for Slave 0, `0x4000_1000` for Slave 1) to `PSELx` select lines.
2. **Glitch-Free Integrated Clock Gating (ICG) Cell:** Negative-level latch + AND gate structure (`latch_en` sampled when `CLK = 0`) preventing runt clock pulses or glitches on `PCLK_gated[0:3]` .
3. **AMBA Handshake FSM (`IDLE -> SETUP -> ACCESS`):** Full compliance with `PSEL` , `PENABLE` , `PWRITE` , `PREADY` , and `PSLVERR` timing rules.
4. **Hardware Bus Watchdog:** Configurable cycle counter (e.g., 16 or 32 cycles) that asserts `PSLVERR = 1` and terminates the transaction if a slow or faulty slave fails to assert `PREADY` .

TIER A: JUNIOR DIVISION (1ST & 2ND YEAR UG)

- Synthesizable APB Master-to-4-Slave interconnect with memory-mapped address decoding, glitch-free clock gating per slave, and working read/write simulation waveforms.

TIER B: SENIOR DIVISION (3RD & 4TH YEAR UG / PG)

- AXI4-Lite-to-APB protocol bridge (converting decoupled AXI read/write channels into APB transfers) OR 2-Master arbitrated APB interconnect with bus watchdog timeout recovery and synthesized dynamic power savings comparison (Ungated vs. Clock-Gated).

SLIDE 5 **MANDATORY PROOF-OF-COMPETENCE ARTIFACT (FOR ROUND 1 PPT)**

Your Slide 5 MUST include: **(a)** The memory-map address decoding table for all slaves, **(b)** The schematic/RTL of your glitch-free Latch+AND clock-gating cell explaining why a bare AND gate causes clock glitches, and **(c)** An APB timing diagram showing `PCLK_gated` waking up only during `PSEL=1` .

TRACK 01: CORE ASIC & CUSTOM SILICON ARCHITECTURE (VLSI-A)

VTX-T1-05 Side-Channel Resistant Cryptographic Datapath (Masked Cipher)

DOMAIN TRACK	**Track 01: Core ASIC & Custom Silicon Architecture (VLSI-A)**
ORGANIZING HOST	**Department of EE - VDT, Sri Shakthi Institute of Engineering and Technology**
CATEGORY	**Hardware Security, Cryptographic Silicon & Side-Channel Countermeasures**

1. BACKGROUND & INDUSTRY CONTEXT — REAL-WORLD CHALLENGE

Even when an encryption algorithm (such as AES-128, PRESENT-80, or Post-Quantum NTT) is mathematically unbreakable, physical silicon implementations leak their secret keys through **Side-Channel Attacks (SCA)**. By placing a resistor on the chip's V_{DD} power pin and measuring micro-ampere current spikes with an oscilloscope (**Differential Power Analysis - DPA**), an attacker can deduce the secret key within minutes because standard CMOS logic consumes power proportional to the Hamming distance ($0 \rightarrow 1$ transitions) of the secret data.

2. PROBLEM DESCRIPTION & SCOPE OF INNOVATION — TECHNICAL OBJECTIVE

Teams must design a hardware cryptographic datapath (e.g., an AES S-Box/SubBytes engine, PRESENT lightweight block cipher, or Simon/Speck core) fortified with **Boolean Masking / Power-Flattening Countermeasures**. By splitting sensitive internal state variables into random shares ($x = x_0 \oplus m$, where m is a fresh pseudo-random mask generated on-chip), the instantaneous switching activity of the circuit becomes statistically independent of the secret key.

3. EXPECTED SYSTEM ARCHITECTURE & FUNCTIONAL SPECIFICATIONS — DELIVERABLES

1. **Baseline Cryptographic Core:** Synthesizable RTL encryption datapath with standard key-schedule and round transformation logic.
2. **On-Chip PRNG Mask Generator:** High-entropy LFSR / Cellular Automata PRNG supplying fresh random mask bits on every clock cycle.
3. **First-Order Masked Non-Linear Layer (S-Box / AND Gate):** Implementation of masked logic gates (e.g., Trichina AND gate or Domain-Oriented Masking with pipeline synchronization flip-flops to prevent glitch leakage).
4. **Toggle-Count / Hamming-Weight Analyzer Testbench:** A verification module that counts total bit-flips (`toggle_count`) per encryption round across different keys to prove reduced correlation between key bits and switching activity.

TIER A: JUNIOR DIVISION (1ST & 2ND YEAR UG)

- Synthesizable RTL of a lightweight cipher (PRESENT-80 or 4-bit/8-bit S-Box datapath) with basic XOR input/output masking and a self-checking encryption/decryption testbench.

TIER B: SENIOR DIVISION (3RD & 4TH YEAR UG / PG)

- First-order Boolean-masked AES S-Box or lightweight cipher with pipelined register barriers (preventing combinational glitch un-masking) and quantitative comparison of switching activity / power reports between the Unmasked vs. Masked RTL.

SLIDE 5 MANDATORY PROOF-OF-COMPETENCE ARTIFACT (FOR ROUND 1 PPT)

Your Slide 5 MUST include: **(a)** The mathematical Boolean masking equation showing how non-linear operations ($A \cdot B$) are computed on masked shares without recombining the secret, and **(b)** Gate-level schematic or RTL interface of the masked S-Box/cipher block.

TRACK 02: FPGA ACCELERATION & RECONFIGURABLE HARDWARE (VLSI-B)

VTX-T2-01 Real-Time 2D Sobel & Canny Edge-Detection Streaming Pipeline

DOMAIN TRACK	**Track 02: FPGA Acceleration & Reconfigurable Hardware (VLSI-B)**
ORGANIZING HOST	**Department of EE - VDT, Sri Shakthi Institute of Engineering and Technology**
CATEGORY	**FPGA Vision Acceleration, Streaming DSP & RTL Image Processing**

1. BACKGROUND & INDUSTRY CONTEXT — REAL-WORLD CHALLENGE

Autonomous drones, high-speed industrial sorting cameras, and ADAS lane-keeping systems require real-time edge detection at 60 to 120 frames per second (FPS). Executing 2D spatial convolutions in software on an embedded CPU introduces massive frame latency and DRAM bandwidth bottlenecks because every pixel requires fetching a 3×3 neighborhood from external memory.

2. PROBLEM DESCRIPTION & SCOPE OF INNOVATION — TECHNICAL OBJECTIVE

Teams must architect a **Zero-Frame-Buffer Streaming 2D Edge-Detection Hardware Accelerator** in Verilog/SystemVerilog/VHDL. Instead of storing an entire image in external RAM, the hardware must ingest 1 grayscale pixel per clock cycle via an AXI-Stream / valid-ready interface, buffer exactly 2 previous image rows inside on-chip **Line Buffers (Shift Registers / Dual-Port BRAM)**, compute horizontal (G_x) and vertical (G_y) spatial gradients in parallel using shift-and-add arithmetic, and output a binary or gradient-magnitude edge pixel on every single clock cycle after initial pipeline fill.

3. EXPECTED SYSTEM ARCHITECTURE & FUNCTIONAL SPECIFICATIONS — DELIVERABLES

1. **3x3 Sliding Window Line-Buffer Architecture:** Two cascaded row-length FIFO/BRAM line buffers feeding a 3×3 shift-register pixel matrix (`P00` to `P22`) simultaneously every clock cycle.
2. **Multiplier-Free Pipelined Gradient Engine:** Computation of $G_x = (P_{02} + 2P_{12} + P_{22}) - (P_{00} + 2P_{10} + P_{20})$ and G_y using strictly hardware bit-shifts (`<< 1`) and adders (zero costly multipliers), followed by fast Manhattan magnitude approximation ($|G| \approx |G_x| + |G_y|$).
3. **Dynamic Thresholding & Border Handling:** Programmable hysteresis/threshold comparator with image boundary padding logic.

TIER A: JUNIOR DIVISION (1ST & 2ND YEAR UG)

RTL implementation of the 3×3 sliding window line buffer and Sobel gradient calculator.

Simulation testbench that reads a standard test image converted to a `.hex` / `.mem` file, runs it through the Verilog simulator, and reconstructs the output `.hex` into a visible edge-detected image.

TIER B: SENIOR DIVISION (3RD & 4TH YEAR UG / PG)

- Full streaming pipeline incorporating Gaussian smoothing pre-filter + Sobel gradient + directional Non-Maximum Suppression (Canny pipeline), synthesized for FPGA (Vivado / Quartus) with LUT/BRAM/Fmax utilization reports or live hardware output.

SLIDE 5 **MANDATORY PROOF-OF-COMPETENCE ARTIFACT (FOR ROUND 1 PPT)**

Your Slide 5 MUST include: **(a)** The hardware block diagram of the 2-row Line Buffer feeding the 9-register 3×3 window, **(b)** Pipeline latency calculation (in clock cycles for a $W \times H$ image), and **(c)** Preliminary input-vs-output pixel matrix verification.

TRACK 02: FPGA ACCELERATION & RECONFIGURABLE HARDWARE (VLSI-B)

VTX-T2-02 Custom RISC-V (RV32I) Core with Hardware AI/DSP Instruction Extension

DOMAIN TRACK	**Track 02: FPGA Acceleration & Reconfigurable Hardware (VLSI-B)**
ORGANIZING HOST	**Department of EE - VDT, Sri Shakthi Institute of Engineering and Technology**
CATEGORY	**Processor Architecture, RISC-V ISA Customization & Hardware Accelerators**

1. BACKGROUND & INDUSTRY CONTEXT — REAL-WORLD CHALLENGE

The open-source **RISC-V Instruction Set Architecture (ISA)** has revolutionized custom silicon because it reserves dedicated instruction encoding spaces (`custom-0` opcode `0x0B` and `custom-1` opcode `0x2B`) for domain-specific hardware acceleration. On a standard microcontroller, running a neural network dot-product ($acc \leftarrow acc + (A \times B)$) or quantized activation function takes 15 to 40 clock cycles per element in software. Adding a custom hardware execution unit directly inside the RISC-V pipeline collapses that loop into a **single clock cycle**.

2. PROBLEM DESCRIPTION & SCOPE OF INNOVATION — TECHNICAL OBJECTIVE

Teams must design a synthesizable **32-bit RISC-V (`RV32I`) Processor Core** in RTL and extend its instruction decoder and execution stage with **Custom AI/DSP Instructions**—such as a Packed 8-bit SIMD Multiply-Accumulate (`simd.mac`), Hardware ReLU/Clipping (`nn.relu`), or Bit-Manipulation Dot-Product—demonstrating a quantifiable cycle-count speedup when executing an edge inference or signal-processing kernel.

3. EXPECTED SYSTEM ARCHITECTURE & FUNCTIONAL SPECIFICATIONS — DELIVERABLES

1. **Base RV32I Datapath:** Program Counter (PC), Instruction Memory, 32x32-bit Register File (`x0` hardwired to `0`), Immediate Generator, ALU, Branch Unit, and Data Memory interface.
2. **Custom Opcode Decoder (`0x0B` / `funct3` / `funct7`):** Clean decoding of R-type custom instructions (`custom_op rd, rs1, rs2`) without breaking standard RV32I instructions (`ADD`, `SUB`, `LW`, `SW`, `BEQ`, `JAL`).
3. **Dedicated Custom Execution Unit:** Tightly coupled MAC accumulator or 4-way 8-bit SIMD dot-product engine operating in parallel with the standard ALU.

TIER A: JUNIOR DIVISION (1ST & 2ND YEAR UG)

Single-cycle or 3-stage RV32I subset processor in Verilog capable of executing a machine-code program from instruction memory, including at least 2 custom AI/DSP instructions (`MAC` and `ReLU`).

Simulation waveform comparing total clock cycles required to compute an 8-element vector dot-product using standard `ADD`/loop instructions vs. your custom instruction.

TIER B: SENIOR DIVISION (3RD & 4TH YEAR UG / PG)

- 5-stage pipelined RV32I core (`IF -> ID -> EX -> MEM -> WB`) with data-forwarding unit, load-use hazard stall logic, packed 4x8-bit INT8 SIMD dot-product unit, and FPGA resource/Fmax synthesis report.

SLIDE 5 **MANDATORY PROOF-OF-COMPETENCE ARTIFACT (FOR ROUND 1 PPT)**

Your Slide 5 MUST include: **(a)** The exact 32-bit instruction encoding bit-field (`funct7 | rs2 | rs1 | funct3 | rd | opcode`) for your custom instructions, **(b)** Your processor datapath diagram highlighting the custom execution path, and **(c)** A cycle-count comparison table.

TRACK 02: FPGA ACCELERATION & RECONFIGURABLE HARDWARE (VLSI-B)

VTX-T2-03 Hardware-Accelerated Wire-Speed Network Packet Firewall

DOMAIN TRACK	**Track 02: FPGA Acceleration & Reconfigurable Hardware (VLSI-B)**
ORGANIZING HOST	**Department of EE - VDT, Sri Shakthi Institute of Engineering and Technology**
CATEGORY	**High-Speed Networking Hardware, FPGA Security & Wire-Speed Parsing**

1. BACKGROUND & INDUSTRY CONTEXT

REAL-WORLD CHALLENGE

In data centers, industrial SCADA networks, and 5G edge gateways, software firewalls running on general-purpose OS kernels suffer from interrupt storms and packet drops under DDoS floods or microsecond-sensitive trading traffic. FPGA-based **SmartNICs and Hardware Firewalls** inspect packet headers and payload signatures directly on the wire at 1 byte (or 64 bits) per clock cycle with deterministic nanosecond latency.

2. PROBLEM DESCRIPTION & SCOPE OF INNOVATION

TECHNICAL OBJECTIVE

Teams must design a **Wire-Speed Hardware Packet Parser & Deep-Packet Inspection (DPI) Firewall Engine** in RTL. As raw network frames stream in over an 8-bit or 32-bit bus (`rx_data`, `rx_valid`, `rx_sof`, `rx_eof`), the hardware must parse the Ethernet, IPv4, and TCP/UDP header fields on the fly, match them against a programmable Access Control List (ACL) table AND a parallel payload signature scanner, and assert `drop_packet = 1` or `forward_packet = 1` before the packet finishes exiting the pipeline.

3. EXPECTED SYSTEM ARCHITECTURE & FUNCTIONAL SPECIFICATIONS

DELIVERABLES

1. **Cut-Through Streaming Header Parser FSM:** Extracts `Source MAC`, `EtherType`, `Source IP`, `Destination IP`, `Protocol (TCP=0x06 / UDP=0x11)`, `Source Port`, and `Destination Port` at exact byte offsets without buffering the entire frame in slow memory.
2. **Parallel Rule-Matching Engine (TCAM / Hash / Comparator Array):** Simultaneously compares extracted 5-tuple header fields against at least 8 configurable security rules (supporting wildcard/subnet masks).
3. **Payload Signature Scanner:** Shift-register or deterministic finite automaton (DFA / Bloom filter) that scans packet payloads for forbidden malicious byte sequences in a single pass.

TIER A: JUNIOR DIVISION (1ST & 2ND YEAR UG)

- Synthesizable RTL streaming parser that decodes simplified IPv4/UDP packet headers from an 8-bit/32-bit input stream, checks against 4 programmable blacklist rules (blocked IP, blocked port, or bad checksum), and asserts `drop_packet` with a telemetry counter.

TIER B: SENIOR DIVISION (3RD & 4TH YEAR UG / PG)

- 32-bit/64-bit AXI-Stream pipelined firewall with Ternary Content-Addressable Memory (TCAM) wildcard rule matching, multi-pattern payload signature detection, and DDoS rate-limiting token bucket per IP, with FPGA synthesis timing reports.

SLIDE 5 **MANDATORY PROOF-OF-COMPETENCE ARTIFACT (FOR ROUND 1 PPT)**

Your Slide 5 MUST include: **(a)** Your byte-offset parsing state machine diagram (showing exact byte indices for Ethernet/IPv4/UDP fields), and **(b)** Architecture of your parallel rule-lookup table.

TRACK 02: FPGA ACCELERATION & RECONFIGURABLE HARDWARE (VLSI-B)

VTX-T2-04 Low-Power MFCC Audio Feature Extractor for Always-On Wake-Word

DOMAIN TRACK	**Track 02: FPGA Acceleration & Reconfigurable Hardware (VLSI-B)**
ORGANIZING HOST	**Department of EE - VDT, Sri Shakthi Institute of Engineering and Technology**
CATEGORY	**Hardware DSP, Fixed-Point Arithmetic & Ultra-Low-Power Audio Silicon**

1. BACKGROUND & INDUSTRY CONTEXT — REAL-WORLD CHALLENGE

Smart glasses, hearing aids, and voice-activated industrial helmets cannot afford to run a main application processor continuously at 100 mW just to listen for a wake-word. Instead, an **Always-On Hardware Audio Front-End** consumes microwatts by first checking acoustic voice activity (VAD) and extracting **Mel-Frequency Cepstral Coefficients (MFCC) / Spectral Energy Bands** using dedicated fixed-point hardware DSP blocks.

2. PROBLEM DESCRIPTION & SCOPE OF INNOVATION — TECHNICAL OBJECTIVE

Teams must design a **Fixed-Point Hardware Audio Feature Extraction Pipeline** in RTL/FPGA that accepts serial digital microphone samples (I2S / 16-bit PCM at 8 kHz or 16 kHz), suppresses DC offset and high-frequency aliasing via a digital pre-emphasis filter, detects active human speech via a Voice Activity Detector (VAD), and computes spectral band energies to trigger an interrupt (`wake_irq`) only when a target acoustic pattern is present.

3. EXPECTED SYSTEM ARCHITECTURE & FUNCTIONAL SPECIFICATIONS — DELIVERABLES

1. **I2S / PCM Serial-to-Parallel Receiver:** Converts serial digital audio into 16-bit signed fixed-point samples (`Q1.15`).
2. **Multiplier-Efficient Pre-Emphasis & Windowing:** First-order high-pass filter `y[n] = x[n] - \alpha x[n-1]` (where `\alpha \approx 0.9375 = 1 - 2^{-4}`, implemented via shift-and-subtract) followed by Hamming window LUT multiplication.
3. **Hardware Spectral / VAD Engine:** Short-Time Energy (STE) + Zero-Crossing Rate (ZCR) gated wake-up stage feeding an 8-point/16-point Radix-2 FFT butterfly or Mel-band filterbank accumulator.

TIER A: JUNIOR DIVISION (1ST & 2ND YEAR UG)

Synthesizable fixed-point RTL implementing the Pre-Emphasis filter, Short-Time Energy (STE) accumulator, and Zero-Crossing Rate (ZCR) Voice Activity Detector that automatically gates downstream clock logic during silence.

Simulation testbench feeding a real `.wav` audio sample (converted to hex) containing silence, noise, and a spoken command.

TIER B: SENIOR DIVISION (3RD & 4TH YEAR UG / PG)

- Complete pipelined fixed-point FFT butterfly + Mel-filterbank log-energy extractor with quantization bit-width error analysis (Floating-point Python golden reference vs. Fixed-point RTL output) and FPGA DSP-slice utilization report.

SLIDE 5 **MANDATORY PROOF-OF-COMPETENCE ARTIFACT (FOR ROUND 1 PPT)**

Your Slide 5 MUST include: **(a)** Fixed-point representation format (`Qm.n` bit-width allocation at each pipeline stage to prevent overflow), and **(b)** Block diagram of the shift-and-add pre-emphasis / FFT butterfly stage.

TRACK 02: FPGA ACCELERATION & RECONFIGURABLE HARDWARE (VLSI-B)

VTX-T2-05 Multi-Channel High-Resolution PWM & Dead-Time Motor Controller

DOMAIN TRACK	**Track 02: FPGA Acceleration & Reconfigurable Hardware (VLSI-B)**
ORGANIZING HOST	**Department of EE - VDT, Sri Shakthi Institute of Engineering and Technology**
CATEGORY	**Power Electronics Control Hardware, EV Inverters & Real-Time FPGA Control**

1. BACKGROUND & INDUSTRY CONTEXT — REAL-WORLD CHALLENGE

In Electric Vehicle (EV) 3-phase traction inverters and multi-axis surgical robotics, upper and lower half-bridge power MOSFETs/IGBTs switch hundreds of amperes. If the High-Side switch (`PWM_H`) and Low-Side switch (`PWM_L`) turn on simultaneously for even 50 nanoseconds due to software interrupt latency, a catastrophic **shoot-through short circuit** destroys the inverter. Furthermore, software-driven PID loops cannot match the sub-microsecond deterministic feedback response of hardware FPGA controllers.

2. PROBLEM DESCRIPTION & SCOPE OF INNOVATION — TECHNICAL OBJECTIVE

Teams must design a **Deterministic Hardware 3-Phase / Multi-Channel PWM Controller with Programmable Dead-Time Insertion, Quadrature Encoder Feedback, and Sub-Cycle Emergency Fault Shutdown** in RTL/FPGA.

3. EXPECTED SYSTEM ARCHITECTURE & FUNCTIONAL SPECIFICATIONS — DELIVERABLES

1. **Center-Aligned Up/Down Triangle Carrier Generator:** Programmable frequency prescaler and 12-bit/16-bit period counter supporting symmetric center-aligned PWM (reducing harmonic distortion in motors).
2. **Non-Overlapping Dead-Time Insertion Unit:** Hardware rising-edge delay counters guaranteeing that `PWM_H` and `PWM_L` never assert high simultaneously, even if register writes are corrupted.
3. **Zero-Latency Hardware Fault Trip (`fault_trip_n`):** Combinational/asynchronous override latch that forces all 6 inverter outputs into a safe high-impedance/off state within 1 clock cycle of an over-current sensor spike, without waiting for CPU intervention.
4. **Quadrature Rotary Encoder Decoder:** 4x resolution `A/B/Index` pulse counter with direction detection and digital glitch debouncing for closed-loop speed/position feedback.

TIER A: JUNIOR DIVISION (1ST & 2ND YEAR UG)

- Synthesizable RTL for a 3-channel complementary PWM generator (`PWM_H[2:0]`, `PWM_L[2:0]`) with programmable duty-cycle registers, adjustable dead-time clock-cycle counter, and instantaneous emergency brake trip.

TIER B: SENIOR DIVISION (3RD & 4TH YEAR UG / PG)

- Closed-loop hardware controller combining the Quadrature Encoder speed estimator + Fixed-Point Hardware PID Regulator (`u[k] = K_p e[k] + K_i \sum e + K_d \Delta e`) + Sine/Space-Vector PWM (SVPWM) modulator with dead-time protection.

SLIDE 5 **MANDATORY PROOF-OF-COMPETENCE ARTIFACT (FOR ROUND 1 PPT)**

Your Slide 5 MUST include: **(a)** A timing waveform diagram illustrating the exact dead-time margin (`T_{dead}`) between `PWM_H` falling edge and `PWM_L` rising edge, and **(b)** Gate/RTL schematic of the shoot-through protection interlock.

TRACK 03: EDGE AI, TINYML & COMPUTER VISION

VTX-T3-01 Sub-50ms Quantized Edge Vision System for PCB & Wafer Defect Triage

DOMAIN TRACK	**Track 03: Edge AI, TinyML & Computer Vision**
ORGANIZING HOST	**Department of EE - VDT, Sri Shakthi Institute of Engineering and Technology**
CATEGORY	**Industrial Computer Vision, Edge AI Quantization & Automated Optical Inspection**

1. BACKGROUND & INDUSTRY CONTEXT — REAL-WORLD CHALLENGE

High-speed Printed Circuit Board (PCB) surface-mount assembly lines and semiconductor wafer inspection stations process several boards per second. Sending high-resolution camera frames to cloud servers introduces network latency (`>300\text{ ms}`) and bandwidth failures, while human visual inspectors suffer from eye fatigue and miss micro-solder bridges, missing tombstoned capacitors, or pinhole copper opens.

2. PROBLEM DESCRIPTION & SCOPE OF INNOVATION — TECHNICAL OBJECTIVE

Teams must build a **100% Offline, Real-Time Edge Automated Optical Inspection (AOI) Vision System** that ingests live camera frames or high-speed test streams, detects and localizes multiple classes of PCB/wafer manufacturing defects (`Missing Hole` , `Mouse Bite` , `Open Circuit` , `Short / Solder Bridge` , `Spur` , `Spurious Copper` / `Missing Component`) in **under 50 milliseconds per frame** on local edge compute, and outputs structured pass/fail triage coordinates.

3. EXPECTED SYSTEM ARCHITECTURE & FUNCTIONAL SPECIFICATIONS — DELIVERABLES

1. **Image Pre-Processing & Alignment Pipeline:** Illumination normalization (CLAHE / homomorphic filtering) and perspective deskewing so the model remains accurate under varying factory lighting angles.
2. **Quantized Lightweight Detection Engine:** Pruned/quantized neural architecture (e.g., YOLOv8-Nano, MobileNetV3-SSD, or hybrid classical-CV + CNN patch classifier) exported to ONNX Runtime, TensorRT, or TFLite (`FP16` / `INT8`).
3. **Real-Time Operator Triage Dashboard:** Local GUI / Web interface displaying live FPS, per-frame inference latency (ms), defect bounding boxes with millimeter/pixel area estimation, and automated defect count logs.

TIER A: JUNIOR DIVISION (1ST & 2ND YEAR UG)

- Functional offline PCB defect detection pipeline (using DeepPCB / HRIPCB or custom dataset) with a live webcam/video-stream dashboard showing bounding boxes, defect class labels, confidence scores, and Pass/Reject tally.

TIER B: SENIOR DIVISION (3RD & 4TH YEAR UG / PG)

- `INT8` / `FP16` quantized model achieving `<50\text{ ms}` CPU/Edge latency, side-by-side benchmark table (FP32 vs. Quantized INT8 model size, mAP, and FPS), synthetic glare/blur robustness mode, and hardware serial/GPIO reject-gate trigger.

SLIDE 5 **MANDATORY PROOF-OF-COMPETENCE ARTIFACT (FOR ROUND 1 PPT)**

Your Slide 5 MUST include: **(a)** Your model layer/quantization pipeline diagram, **(b)** A concrete latency and memory footprint budget table (Model weight size in MB, RAM usage, target FPS), and **(c)** Preliminary precision/recall or sample detection output.

TRACK 03: EDGE AI, TINYML & COMPUTER VISION

VTX-T3-02 Non-Invasive Driver Drowsiness, Gaze & Micro-Sleep Edge Telemetry

DOMAIN TRACK	**Track 03: Edge AI, TinyML & Computer Vision**
ORGANIZING HOST	**Department of EE - VDT, Sri Shakthi Institute of Engineering and Technology**
CATEGORY	**Automotive ADAS, Real-Time Biometric Vision & Multi-Modal Edge AI**

1. BACKGROUND & INDUSTRY CONTEXT — REAL-WORLD CHALLENGE

Highway commercial fleet accidents between 1:00 AM and 5:00 AM are overwhelmingly caused by **micro-sleep episodes** (lasting 2 to 4 seconds) and prolonged off-road gaze distraction. Simple eye-blink counters fail in the real world because they trigger false alarms when a driver squints in bright sunlight, smiles, or checks their side-view mirror intentionally.

2. PROBLEM DESCRIPTION & SCOPE OF INNOVATION — TECHNICAL OBJECTIVE

Teams must build a **Low-False-Positive, Non-Invasive In-Cabin Driver Monitoring System (DMS)** running entirely offline at the edge. Rather than relying on a single brittle threshold, the system must fuse **Eye Aspect Ratio (EAR) / PERCLOS (Percentage of Eye Closure over time)**, **Mouth Aspect Ratio (MAR) yawning frequency**, and **3D Head-Pose Yaw/Pitch angles** to compute a continuous **Composite Fatigue Risk Score (0–100)**, triggering graduated alerts (`Stage 1: Advisory -> Stage 2: Seat/Buzzer Warning -> Stage 3: Emergency Hazard Braking Telemetry`).

3. EXPECTED SYSTEM ARCHITECTURE & FUNCTIONAL SPECIFICATIONS — DELIVERABLES

1. **Sub-Frame Facial Landmark & Iris/Head-Pose Extractor:** Ultra-fast 68-point or 468-point facial mesh tracking operating at `\ge 20\text{ FPS}` on standard CPU/edge hardware without GPU cloud offloading.
2. **Temporal State Machine (Anti-Blink Filter):** Distinguishes normal 200ms human blinks from dangerous `>1.5\text{ s}` micro-sleep closures and tracks 60-second rolling PERCLOS windows.
3. **Low-Light / Occlusion Resilience:** Dynamic gamma correction for night-driving simulation and sunglasses/hand-occlusion detection alerts.

TIER A: JUNIOR DIVISION (1ST & 2ND YEAR UG)

- Real-time webcam driver monitoring prototype computing live EAR (eye closure), MAR (yawn detection), and head-tilt alerts with a real-time graphical dashboard and audio buzzer alarm.

TIER B: SENIOR DIVISION (3RD & 4TH YEAR UG / PG)

- Multi-modal fusion combining vision telemetry (PERCLOS + 3D Euler head pose) with simulated or physical vehicle telemetry (steering wheel micro-corrections / accelerometer jerk / heart-rate sensor), plus per-driver auto-calibration during the first 10 seconds to handle different eye shapes.

SLIDE 5 **MANDATORY PROOF-OF-COMPETENCE ARTIFACT (FOR ROUND 1 PPT)**

Your Slide 5 MUST include: **(a)** The mathematical equations for EAR, MAR, and your Composite Fatigue Score fusion weights, and **(b)** The temporal state-transition diagram showing how normal blinks and mirror-checks are filtered out.

TRACK 03: EDGE AI, TINYML & COMPUTER VISION

VTX-T3-03 Acoustic Anomaly & Predictive Bearing Fault Detector for Motors

DOMAIN TRACK	**Track 03: Edge AI, TinyML & Computer Vision**
ORGANIZING HOST	**Department of EE - VDT, Sri Shakthi Institute of Engineering and Technology**
CATEGORY	**Predictive Maintenance, TinyML Audio/Vibration DSP & Industrial AI**

1. BACKGROUND & INDUSTRY CONTEXT

REAL-WORLD CHALLENGE

In semiconductor fab cleanrooms, CNC machining plants, and textile mills across Coimbatore, an unexpected induction motor or coolant pump failure halts an entire production line, costing lakhs of rupees per hour. Days before a motor physically seizes, microscopic pitting in its ball bearings or rotor eccentricity generates distinct **acoustic and vibration harmonic signatures**.

2. PROBLEM DESCRIPTION & SCOPE OF INNOVATION

TECHNICAL OBJECTIVE

Teams must build an **Edge Acoustic & Vibration Anomaly Detection System** that listens to real-time microphone audio (or accelerometer vibration streams) from rotating machinery, transforms the time-domain signal into frequency-domain spectrograms (STFT / Mel-Spectrogram / Wavelet), and uses an **Unsupervised Autoencoder or Lightweight 1D/2D CNN Classifier** to detect early bearing wear, shaft misalignment, unbalanced loads, or cavitation—even in the presence of ambient factory chatter.

3. EXPECTED SYSTEM ARCHITECTURE & FUNCTIONAL SPECIFICATIONS

DELIVERABLES

1. **Real-Time Signal Conditioning & Spectral Pipeline:** High-pass noise floor removal, Hanning windowing, and FFT / Mel-Spectrogram generation updated every 100ms–250ms.
2. **Dual-Mode Diagnostic Engine:**

Anomaly Detection Score:* Reconstruction error `\|X - \hat{X}\|^2` (for unseen anomalies) OR multi-class fault probability (`Healthy`, `Inner/Outer Bearing Race Fault`, `Rotor Imbalance`, `Misalignment`).

1. **Remaining Useful Life (RUL) / Maintenance Dashboard:** Live waterfall spectrogram display, harmonic peak tracker, and automated maintenance ticket alert when anomaly thresholds persist across `N` consecutive windows.

TIER A: JUNIOR DIVISION (1ST & 2ND YEAR UG)

- Real-time audio diagnostic dashboard capable of ingesting live microphone input or MIMII / CWRU bearing dataset audio files, plotting live FFT/spectrograms, and accurately classifying healthy vs. faulty machine states.

TIER B: SENIOR DIVISION (3RD & 4TH YEAR UG / PG)

- TinyML model (`< 150 KB` footprint) capable of running on an edge microcontroller/Raspberry Pi/laptop CPU with active background-noise cancellation (SNR stress testing at `0\text{ dB}` to `-6\text{ dB}`) and physical vibration/audio hardware rig demonstration.

SLIDE 5 **MANDATORY PROOF-OF-COMPETENCE ARTIFACT (FOR ROUND 1 PPT)**

Your Slide 5 MUST include: **(a)** Your signal-processing pipeline parameters (Sampling rate `f_s`, FFT window size, hop length, frequency resolution `\Delta f`), and **(b)** A sample frequency spectrum / spectrogram comparison between a healthy and faulty bearing state.

TRACK 03: EDGE AI, TINYML & COMPUTER VISION

VTX-T3-04 Offline Edge-SLM & RAG Assistant for Datasheet & Pinout Triage

DOMAIN TRACK	**Track 03: Edge AI, TinyML & Computer Vision**
ORGANIZING HOST	**Department of EE - VDT, Sri Shakthi Institute of Engineering and Technology**
CATEGORY	**Offline Small Language Models (SLMs), Hardware RAG & Engineering Productivity**

1. BACKGROUND & INDUSTRY CONTEXT

REAL-WORLD CHALLENGE

Hardware and VLSI engineers spend hours cross-referencing 800-page microcontroller reference manuals (e.g., STM32, ESP32, RISC-V SoCs) and IC datasheets to find exact register bit-fields, absolute maximum voltage ratings, I2C slave addresses, and pin multiplexing tables. Cloud chatbots cannot be used inside air-gapped defense/semiconductor labs due to IP security rules, and generic LLMs frequently hallucinate wrong pin numbers—which can physically fry a board!

2. PROBLEM DESCRIPTION & SCOPE OF INNOVATION

TECHNICAL OBJECTIVE

Teams must build a **100% Offline, Air-Gapped Hardware Engineering RAG (Retrieval-Augmented Generation) Assistant** powered by a quantized local Small Language Model (SLM) and a table-aware PDF parser. When an engineer drops one or more IC datasheets/manuals into the tool and asks a hardware query (e.g., "*What are the control bits for Timer1 PWM mode and what is the max current on GPIO pin PA5?*"), the system must return the exact technical answer **strictly grounded in the datasheet with the exact Page Number, Section Title, and Table/Register snippet**, refusing to guess when data is absent.

3. EXPECTED SYSTEM ARCHITECTURE & FUNCTIONAL SPECIFICATIONS

DELIVERABLES

1. **Table & Register-Map Aware Ingestion Engine:** Standard text chunkers destroy datasheet pinout tables and register bit-maps. The pipeline must preserve markdown tables, pinout grids, and electrical characteristic rows intact during vector/hybrid indexing.
2. **100% Local Inference Engine (Zero External API Keys):** Must execute entirely offline with Wi-Fi turned OFF using quantized local models (e.g., Llama-3.2-1B/3B, Qwen2.5-Coder-1.5B/3B, Phi-3.5-Mini via Ollama / llama.cpp / ONNX) paired with hybrid BM25 keyword + dense vector retrieval.
3. **Deterministic Citation & Pin-Conflict Checker:** Every response must display clickable/highlighted source chunks from the original PDF page, plus a structured JSON/Verilog/C header generator for register definitions.

TIER A: JUNIOR DIVISION (1ST & 2ND YEAR UG)

- Fully offline local RAG desktop/web application that ingests multi-page component datasheets, answers pinout and electrical spec queries with exact page citations, and passes the "Airplane Mode (Wi-Fi Off)" live audit.

TIER B: SENIOR DIVISION (3RD & 4TH YEAR UG / PG)

- Hybrid search (Exact Hex/Part-Number BM25 + Semantic Vector Search), automated generation of C `.h` register macros or Verilog module instantiation templates directly from datasheet tables, and cross-datasheet compatibility checking (e.g., flagging if a 5V sensor output is wired to a non-5V-tolerant 3.3V MCU pin).

SLIDE 5 **MANDATORY PROOF-OF-COMPETENCE ARTIFACT (FOR ROUND 1 PPT)**

Your Slide 5 MUST include: **(a)** Your table-preserving chunking and hybrid retrieval architecture diagram, **(b)** Exact local model quantization specs (Model name, GGUF/INT4 quantization level, RAM required, tokens/sec on CPU), and **(c)** A sample input datasheet table vs. extracted structured output.

TRACK 03: EDGE AI, TINYML & COMPUTER VISION

VTX-T3-05 Precision Agricultural Drone Imagery Triage for Early Crop Stress

DOMAIN TRACK	**Track 03: Edge AI, TinyML & Computer Vision**
ORGANIZING HOST	**Department of EE - VDT, Sri Shakthi Institute of Engineering and Technology**
CATEGORY	**Agri-Tech Edge Vision, Spectral Segmentation & Low-Power Drone Compute**

1. BACKGROUND & INDUSTRY CONTEXT — REAL-WORLD CHALLENGE

Farmers across Tamil Nadu lose up to 30% of their crop yield to localized fungal blights, pest infestations, and nitrogen/water deficiency because blanket pesticide spraying is applied too late or wasted across healthy acreage. Agricultural drones capture aerial foliage imagery, but rural farms lack high-speed internet to upload gigabytes of raw video to the cloud—requiring **on-drone or field-edge real-time triage**.

2. PROBLEM DESCRIPTION & SCOPE OF INNOVATION — TECHNICAL OBJECTIVE

Teams must design a **Low-Power Edge Vision & Spectral Triage Pipeline** that processes aerial or close-canopy crop imagery offline in the field, segments healthy foliage from diseased/stressed zones, classifies the specific pathology (`Bacterial Blight`, `Leaf Blast`, `Rust`, `Pest Damage`, `Nutrient/Water Stress`), computes the **Infected Leaf Area Percentage**, and generates an actionable **Grid-Zone Spray/Dosage Map** so farmers only spray affected plots.

3. EXPECTED SYSTEM ARCHITECTURE & FUNCTIONAL SPECIFICATIONS — DELIVERABLES

1. **Vegetation Index & Color-Space Pre-Filter:** Fast RGB/HSV/ExG (Excess Green Index `2G - R - B`) or simulated NIR background masking to strip away soil, rocks, and shadows before running neural inference.
2. **Lightweight Classification & Severity Segmentation:** Pruned/quantized CNN or Vision Transformer (MobileViT / ShuffleNet / YOLOv8-Seg) classifying pathology and calculating lesion-to-leaf area ratio.
3. **Field Action & Prescription Generator:** Offline dashboard mapping each image frame to a farm grid coordinate `(Row, Col)`, outputting exact severity level (`Low <10%`, `Moderate 10-35%`, `Critical >35%`) and recommended localized treatment dosage.

TIER A: JUNIOR DIVISION (1ST & 2ND YEAR UG)

- Offline crop stress classification and leaf-lesion area segmentation pipeline with a clean field dashboard displaying disease class, affected leaf area percentage, and recommended action.

TIER B: SENIOR DIVISION (3RD & 4TH YEAR UG / PG)

- Quantized edge model achieving `\ge 15\text{ FPS}` on CPU/edge board, stitched multi-tile farm grid heat-map generator calculating total pesticide volume saved (in % compared to blanket spraying), and motion-blur/sunlight resilience.

SLIDE 5 **MANDATORY PROOF-OF-COMPETENCE ARTIFACT (FOR ROUND 1 PPT)**

Your Slide 5 MUST include: **(a)** The mathematical equation used for soil/background masking and lesion area percentage calculation, and **(b)** Model architecture, parameter count, and inference latency benchmarks.

TRACK 04: SMART SYSTEMS, CYBER-PHYSICAL IOT & AUTOMATION

VTX-T4-01 Non-Linear Sensor Noise Rejection & Closed-Loop Actuator Stabilization

DOMAIN TRACK	**Track 04: Smart Systems, Cyber-Physical IoT & Automation**
ORGANIZING HOST	**Department of EE - VDT, Sri Shakthi Institute of Engineering and Technology**
CATEGORY	**Embedded Control Systems, Real-Time Digital Signal Filtering & Robotics**

1. BACKGROUND & INDUSTRY CONTEXT — REAL-WORLD CHALLENGE

In industrial automation, autonomous ground vehicles, and chemical level-control tanks, low-cost Ultrasonic, Infrared, ToF, and pressure sensors frequently produce **non-Gaussian impulse noise spikes (glitches)** caused by acoustic multipath reflections, EMI motor switching noise, or surface turbulence. Feeding raw sensor readings—or even simple linear moving-average outputs—directly into a PID actuator loop causes violent servo/valve hunting, mechanical wear, and control instability.

2. PROBLEM DESCRIPTION & SCOPE OF INNOVATION — TECHNICAL OBJECTIVE

Teams must build a **Real-Time Embedded Non-Linear Sensor Filtering & Closed-Loop Actuator Stabilization System** accompanied by a live 60fps telemetry visualizer. The system must sample live sensor data (and support synthetic fault/spike injection for stress testing), strip out severe impulse outliers without introducing phase lag that destabilizes the loop, drive a physical or simulated actuator (servo, motor, or control output) smoothly to a target setpoint, and plot raw vs. filtered vs. actuator signals in real time.

3. EXPECTED SYSTEM ARCHITECTURE & FUNCTIONAL SPECIFICATIONS — DELIVERABLES

1. **Cascaded Non-Linear + Optimal Filter Pipeline:** Combination of a sliding-window **Median / Hampel Outlier Filter** (to reject sharp impulse spikes completely) cascaded with an **Adaptive 1D Kalman Filter or Exponentially Weighted Moving Average (EWMA)** (to smooth Gaussian white noise with minimal phase delay).
2. **Anti-Windup Closed-Loop Controller:** Proportional-Integral-Derivative (PID) controller with integral clamping (anti-windup) and derivative-on-measurement (preventing derivative kick on setpoint jumps).
3. **Live Dual-Trace Telemetry & Fault-Injection Console:** Real-time serial/WebSocket dashboard plotting `Raw Noisy Input`, `Filtered Estimate`, `Setpoint`, and `Actuator Output`, alongside a button/switch to inject deliberate noise bursts and measure settling time (`T_s`) and overshoot (%).

TIER A: JUNIOR DIVISION (1ST & 2ND YEAR UG)

- Microcontroller prototype (Arduino / ESP32 / STM32) or real-time hardware-in-the-loop setup reading a physical sensor, running a combined Median + smoothing filter and proportional/PID control loop to stabilize a servo/motor/output, with live serial plotter visualization.

TIER B: SENIOR DIVISION (3RD & 4TH YEAR UG / PG)

- Adaptive 1D Kalman Filter + anti-windup PID with real-time RMSE (Root Mean Square Error) and settling-time telemetry dashboard, plus automatic sensor-disconnect / stuck-at-rail fail-safe fallback mode.

SLIDE 5 **MANDATORY PROOF-OF-COMPETENCE ARTIFACT (FOR ROUND 1 PPT)**

Your Slide 5 MUST include: **(a)** The discrete-time mathematical equations for your filter and anti-windup PID loop, and **(b)** A step-response or spike-rejection plot demonstrating why a Median+Kalman cascade beats a simple moving average on impulse spikes.

TRACK 04: SMART SYSTEMS, CYBER-PHYSICAL IOT & AUTOMATION

VTX-T4-02 EV Battery Management System (BMS) with Thermal Runaway Cutoff

DOMAIN TRACK	**Track 04: Smart Systems, Cyber-Physical IoT & Automation**
ORGANIZING HOST	**Department of EE - VDT, Sri Shakthi Institute of Engineering and Technology**
CATEGORY	**Electric Vehicle (EV) Safety Systems, Embedded Power & Predictive Telemetry**

1. BACKGROUND & INDUSTRY CONTEXT — REAL-WORLD CHALLENGE

Electric two-wheelers and energy storage packs in Indian summer ambient temperatures (`42^\circ\text{C}+`) face severe safety risks from **Lithium-Ion Thermal Runaway**. Standard fuses and basic BMS boards only trip *after* a cell exceeds `65^\circ\text{C}` —by which point exothermic chemical breakdown has already become unstoppable. A truly safe BMS must monitor the **rate of temperature rise (`dT/dt`)**, per-cell voltage divergence, and internal resistance degradation to isolate a failing cell minutes before ignition.

2. PROBLEM DESCRIPTION & SCOPE OF INNOVATION — TECHNICAL OBJECTIVE

Teams must design and prototype an **Intelligent Multi-Cell EV Battery Management & Predictive Thermal Runaway Protection Node**. The system must continuously monitor individual cell voltages (`V_1 \dots V_N`), pack current (`I_{pack}`), and multi-point cell temperatures (`T_1 \dots T_N`), execute real-time State-of-Charge (SoC) and passive/active cell-balancing logic, detect rapid thermal gradients (`dT/dt > \text{threshold}`) or over-current shorts, and physically trip a solid-state MOSFET/relay contactor within milliseconds while broadcasting black-box fault logs.

3. EXPECTED SYSTEM ARCHITECTURE & FUNCTIONAL SPECIFICATIONS — DELIVERABLES

1. **Multi-Channel Sensing & Fault Simulation Array:** Sensing of at least 3 to 4 series cell banks (using real Li-ion cells or precision variable voltage/thermistor dividers for safe fault injection in the arena).
2. **Predictive Thermal & Electrical Safety Engine:**

- Monitors Over-Voltage (`>4.25\text{V}`), Under-Voltage (`<2.80\text{V}`), Over-Current, Absolute Over-Temperature, AND **Rate-of-Thermal-Rise (`dT/dt` in `^\circ\text{C/sec}`)**.

1. **Cell Balancing & Contactor Cutoff Hardware:** Bleed-resistor MOSFET switches across each cell to equalize high cells during charging, plus a master main-bus cutoff relay/MOSFET and active cooling fan PWM control.

TIER A: JUNIOR DIVISION (1ST & 2ND YEAR UG)

- Working embedded BMS prototype monitoring 3+ cell voltage/temperature channels, displaying live cell voltages and SoC % on an OLED/Web dashboard, and physically tripping a cutoff relay + cooling fan when an over-voltage or thermal spike is induced.

TIER B: SENIOR DIVISION (3RD & 4TH YEAR UG / PG)

- Coulomb-counting + Open-Circuit-Voltage (OCV) fused SoC estimation, automated passive cell-balancing MOSFET switching, predictive `dT/dt` thermal runaway early warning, and non-volatile black-box fault log storage.

SLIDE 5 **MANDATORY PROOF-OF-COMPETENCE ARTIFACT (FOR ROUND 1 PPT)**

Your Slide 5 MUST include: **(a)** Complete circuit schematic of your multi-cell voltage divider/sensing network, balancing bleed switches, and main cutoff driver, and **(b)** Your BMS fault-protection state machine table with exact voltage and `dT/dt` trip thresholds.

TRACK 04: SMART SYSTEMS, CYBER-PHYSICAL IOT & AUTOMATION

VTX-T4-03 Decentralized V2X Highway Black-Spot & Fog Collision Avoidance Mesh

DOMAIN TRACK	**Track 04: Smart Systems, Cyber-Physical IoT & Automation**
ORGANIZING HOST	**Department of EE - VDT, Sri Shakthi Institute of Engineering and Technology**
CATEGORY	**Vehicular Ad-Hoc Networks (V2X), Low-Latency Wireless Mesh & Road Safety**

1. BACKGROUND & INDUSTRY CONTEXT — REAL-WORLD CHALLENGE

Multi-vehicle pileups in dense winter fog and head-on collisions on blind mountain hairpin bends (such as the Nilgiris/Ooty ghat roads near Coimbatore) occur because drivers have less than 15 meters of visual line-of-sight, and cellular 4G/5G networks are either dead in mountain terrain or too high-latency for sub-second emergency braking broadcasts.

2. PROBLEM DESCRIPTION & SCOPE OF INNOVATION — TECHNICAL OBJECTIVE

Teams must build a **Zero-Internet, Decentralized Vehicle-to-Everything (V2V / V2I) Collision Avoidance Mesh Network** using low-latency local wireless protocols (e.g., **ESP-NOW, LoRa, nRF24L01, or IEEE 802.15.4**). When a lead vehicle slams its brakes (high deceleration `-a_x`), skids, or passes a roadside hairpin-bend beacon, the node must broadcast a signed hazard packet that hops across trailing vehicles in **under 25 milliseconds**, alerting approaching drivers *around blind corners or through zero-visibility fog*.

3. EXPECTED SYSTEM ARCHITECTURE & FUNCTIONAL SPECIFICATIONS — DELIVERABLES

1. **Vehicle Node & Roadside Unit (RSU) Hardware:** At least 2 to 3 physical wireless transceiver nodes equipped with IMU accelerometer/gyroscope (for hard-braking/rollover detection), ultrasonic/ToF proximity sensors, and directional visual/audio cockpit alerts.
2. **Broadcast Storm Prevention (TTL & Packet Deduplication):** In a wireless mesh, naive rebroadcasting floods the RF channel. Nodes must implement a lightweight packet header (`Node_ID`, `Sequence_Num`, `Hazard_Type`, `TTL_Hop_Count`, `RSSI_Distance`) with a ring-buffer cache to drop duplicate packets.
3. **Blind-Curve / Hairpin Approach Conflict Resolver:** Roadside beacon logic that detects two vehicles approaching a narrow blind curve from opposite directions simultaneously and warns both drivers with relative proximity severity.

TIER A: JUNIOR DIVISION (1ST & 2ND YEAR UG)

- 2+ physical microcontroller nodes communicating peer-to-peer without any Wi-Fi router or internet connection, demonstrating instant emergency braking/obstacle alert propagation and RSSI distance estimation.

TIER B: SENIOR DIVISION (3RD & 4TH YEAR UG / PG)

- 3-node multi-hop mesh (`Node A -> Relay Node B -> Node C` when A and C are out of direct range) with TTL deduplication, lightweight HMAC/CRC packet authentication against spoofed hazard packets, and a live telemetry radar dashboard showing round-trip latency in milliseconds.

SLIDE 5 **MANDATORY PROOF-OF-COMPETENCE ARTIFACT (FOR ROUND 1 PPT)**

Your Slide 5 MUST include: **(a)** The exact custom wireless byte-frame structure (Header, Payload, TTL, Checksum), and **(b)** Your multi-hop packet relay and duplicate-suppression flowchart.

TRACK 04: SMART SYSTEMS, CYBER-PHYSICAL IOT & AUTOMATION

VTX-T4-04 Smart Grid Micro-Load Shedding & Power-Theft Tamper Detection Node

DOMAIN TRACK	**Track 04: Smart Systems, Cyber-Physical IoT & Automation**
ORGANIZING HOST	**Department of EE - VDT, Sri Shakthi Institute of Engineering and Technology**
CATEGORY	**Smart Grid Energy Systems, Embedded Instrumentation & Cyber-Physical Security**

1. BACKGROUND & INDUSTRY CONTEXT

REAL-WORLD CHALLENGE

Distribution utilities in India face two massive operational challenges: **(1) Peak-Hour Transformer Overloads**, where tripping the main feeder blacks out critical loads (hospitals, oxygen concentrators, server racks) alongside non-essential heavy loads (air conditioners, water heaters), and **(2) Distribution Power Theft**, where illegal hooks bypass the phase line or neutral return path of conventional energy meters.

2. PROBLEM DESCRIPTION & SCOPE OF INNOVATION

TECHNICAL OBJECTIVE

Teams must architect an **Intelligent Smart-Grid Micro-Load Shedding & Differential Tamper-Detection Energy Node**. Instead of cutting off an entire building during a grid overload or tariff spike, the node must dynamically shed lower-priority sub-circuits (`Tier 3: Non-Essential -> Tier 2: Deferrable`) while keeping `Tier 1: Critical Loads` online. Simultaneously, it must monitor **Phase vs. Neutral/Upstream Differential Current (`|I_{phase} - I_{neutral}| > I_{threshold}`)** and magnetic/enclosure tamper sensors to detect bypass hooking in real time.

3. EXPECTED SYSTEM ARCHITECTURE & FUNCTIONAL SPECIFICATIONS

DELIVERABLES

1. **Dual-Point Current & Voltage Sensing:** RMS voltage and dual current sensing (Phase current `I_P` and Neutral/Upstream current `I_N`) to compute Active Power (`P`), Apparent Power (`S`), Power Factor (`\cos\phi`), and differential current leakage.
2. **3-Tier Priority Load-Shedding State Machine:** Controls 3 independent relay/SSR channels (`Priority 1: Critical` , `Priority 2: Normal` , `Priority 3: Heavy Non-Essential`). Automatically sheds Priority 3 (and then Priority 2 with hysteresis recovery) when total current exceeds the dynamic grid cap.
3. **Anti-Tamper Security & Utility Console:** Detects parallel wire bypassing (`I_P \neq I_N`), physical meter cover opening, or external magnetic reed/Hall-effect interference, logging tamper timestamps and alerting the utility dashboard.

TIER A: JUNIOR DIVISION (1ST & 2ND YEAR UG)

- Safe low-voltage AC/DC smart-meter prototype with current/voltage sensors and 3 relay channels demonstrating automatic priority load shedding when load exceeds a threshold, plus bypass tamper detection.

TIER B: SENIOR DIVISION (3RD & 4TH YEAR UG / PG)

- True RMS power/power-factor computation, differential current tamper detection + magnetic/cover tamper interlock, hysteresis auto-reconnection timer (preventing relay chatter), and bi-directional utility command dashboard allowing remote demand-response power capping.

SLIDE 5 **MANDATORY PROOF-OF-COMPETENCE ARTIFACT (FOR ROUND 1 PPT)**

Your Slide 5 MUST include: **(a)** Circuit schematic showing how dual current sensors detect a phase bypass hook across the meter, and **(b)** Your Priority Load-Shedding & Hysteresis Reconnection state diagram.

TRACK 04: SMART SYSTEMS, CYBER-PHYSICAL IOT & AUTOMATION

VTX-T4-05 Wearable Multi-Vital Patient Triage Node with Fall & Arrhythmia Alerting

DOMAIN TRACK	**Track 04: Smart Systems, Cyber-Physical IoT & Automation**
ORGANIZING HOST	**Department of EE - VDT, Sri Shakthi Institute of Engineering and Technology**
CATEGORY	**Biomedical Electronics, Low-Power Wearables & Clinical IoT Triage**

1. BACKGROUND & INDUSTRY CONTEXT — REAL-WORLD CHALLENGE

In crowded general hospital wards and post-operative recovery rooms, 1 nurse often monitors 25 to 40 patients whose vitals are checked manually only once every 4 to 6 hours. Sudden clinical deterioration—such as silent hypoxia (`SpO_2 < 90\%`), tachycardia, or an unassisted bathroom fall—goes unnoticed during that gap. However, standard wearable PPG pulse sensors produce massive false alarms whenever a patient simply moves their hand (**motion artifacts**).

2. PROBLEM DESCRIPTION & SCOPE OF INNOVATION — TECHNICAL OBJECTIVE

Teams must build a **Motion-Artifact-Resistant Wearable Multi-Vital Triage Node & Central Nurse-Station Priority Dashboard**. The wearable must continuously sample photoplethysmography (`PPG` for Heart Rate & `SpO_2`), skin/body temperature, and 3-axis accelerometer/gyroscope kinematics (`IMU`), use IMU motion data to suppress false PPG heart-rate spikes during patient movement, detect hard falls (`Free-Fall -> Impact Spike -> Post-Fall Inactivity`), and compute a clinical **Modified Early Warning Score (MEWS)** to rank patients by urgency on a live ward screen.

3. EXPECTED SYSTEM ARCHITECTURE & FUNCTIONAL SPECIFICATIONS — DELIVERABLES

1. **Multi-Sensor Acquisition & Motion-Gated PPG Filter:** Correlates 3-axis accelerometer magnitude `\|a\| = \sqrt{a_x^2 + a_y^2 + a_z^2}` with the optical PPG waveform to reject or flag corrupted heart-rate readings during high hand motion.
2. **3-Phase Biomechanical Fall-Detection Algorithm:** Eliminates false fall alerts (such as clapping or sitting down quickly) by verifying three sequential phases: **(1)** Weightlessness drop (`\|a\| < 0.4g`), **(2)** Impact peak (`\|a\| > 2.5g`), and **(3)** Horizontal tilt change + 5-second post-impact stillness.
3. **Offline Ring-Buffer & Nurse Triage Console:** Stores vital logs locally in flash/RAM if Wi-Fi drops and automatically sorts patient cards on the nurse dashboard so `CRITICAL (Red)` patients jump to the top of the screen with audible alerts.

TIER A: JUNIOR DIVISION (1ST & 2ND YEAR UG)

- Working wearable hardware prototype (ESP32 / Arduino + MAX30102/Pulse + MPU6050 + Temp sensor) streaming live Heart Rate, `SpO_2` , Temperature, and Fall Detection alerts to a clean nurse-station web dashboard.

TIER B: SENIOR DIVISION (3RD & 4TH YEAR UG / PG)

- Motion-artifact cancellation/gating using IMU correlation, 3-phase biomechanical fall state machine (proven not to false-trigger on a simple wrist shake), clinical MEWS score computation, and store-and-forward Wi-Fi disconnect recovery.

SLIDE 5 **MANDATORY PROOF-OF-COMPETENCE ARTIFACT (FOR ROUND 1 PPT)**

Your Slide 5 MUST include: **(a)** The 3-phase fall-detection acceleration threshold state machine, **(b)** Your MEWS clinical scoring lookup table, and **(c)** Hardware wiring/power-budget block diagram.

TRACK 05: OPEN SILICON & HARDWARE INNOVATION

VTX-T5-01 Open Innovation in Assistive Tech & Neuro-Rehabilitation Hardware

DOMAIN TRACK	**Track 05: Open Silicon & Hardware Innovation**
ORGANIZING HOST	**Department of EE - VDT, Sri Shakthi Institute of Engineering and Technology**
CATEGORY	**Assistive Hardware, Biomedical Devices, Human-Machine Interfaces & Edge AI**

1. BACKGROUND & INDUSTRY CONTEXT

REAL-WORLD CHALLENGE

Over 26 million people in India live with visual, auditory, vocal, motor, or neurological impairments, yet imported assistive hardware (such as myoelectric prosthetics, refreshable Braille displays, or eye-tracked communication tablets) costs between ₹1.5 Lakh and ₹10 Lakh—putting them completely out of reach for common families.

2. PROBLEM DESCRIPTION & SCOPE OF INNOVATION

TECHNICAL OBJECTIVE

Teams are invited to pitch and build an **original, low-cost, high-reliability hardware, embedded, Edge-AI, or FPGA/VLSI-accelerated assistive system** that restores independence, mobility, communication, or rehabilitation tracking for differently-abled individuals.

Representative Focus Areas (Teams may propose these or any original assistive hardware concept):

Real-time two-way Sign-Language / Gesture-to-Speech wearable glove or edge-vision translator.

EMG / EEG / Eye-Gaze controlled wheelchair navigation or smart-home actuation interface with obstacle safety override.

Haptic / Ultrasonic spatial spatial-navigation wearable belt or smart glasses for the visually impaired (detecting both ground potholes and head-level overhanging obstacles).

- Quantitative stroke neuro-rehabilitation exoskeleton/glove tracking finger range-of-motion and tremor reduction.

TIER A: JUNIOR DIVISION (1ST & 2ND YEAR UG)

Working hardware/sensor prototype demonstrating core input-to-output assistive functionality with sub-second response time and a clear Bill of Materials (BOM).

TIER B: SENIOR DIVISION (3RD & 4TH YEAR UG / PG)

Complete ergonomic prototype with calibration for individual users, false-trigger safety interlocks, battery-life/power optimization, and quantitative accuracy/latency benchmarking.

SLIDE 5 **MANDATORY PROOF-OF-COMPETENCE ARTIFACT (FOR ROUND 1 PPT)**

Your Slide 5 MUST include: **(a)** Full hardware schematic or system block diagram with exact sensor/actuator part numbers, **(b)** Detailed Bill of Materials (BOM) proving affordability, and **(c)** Signal-processing or control-loop latency specification.

TRACK 05: OPEN SILICON & HARDWARE INNOVATION

VTX-T5-02 Open Innovation in Sustainable Tech, Water & Climate Hardware

DOMAIN TRACK	**Track 05: Open Silicon & Hardware Innovation**
ORGANIZING HOST	**Department of EE - VDT, Sri Shakthi Institute of Engineering and Technology**
CATEGORY	**Clean-Tech Hardware, Smart Water Grid, Renewable Energy & Environmental Sensing**

1. BACKGROUND & INDUSTRY CONTEXT

REAL-WORLD CHALLENGE

Municipalities, industrial corridors, and agricultural belts face acute water pipeline losses (up to 40% Non-Revenue Water lost to underground leaks), untreated industrial effluent discharge into rivers, and inefficient solar photovoltaic energy harvesting under partial cloud shading.

2. PROBLEM DESCRIPTION & SCOPE OF INNOVATION

TECHNICAL OBJECTIVE

Teams are invited to design and build an **original sensor-driven hardware, power-electronics, IoT, or custom silicon/FPGA system** addressing water conservation, contamination detection, renewable energy optimization, or environmental climate resilience.

Representative Focus Areas (Teams may propose these or any original sustainability hardware concept):

Differential flow & acoustic pressure-transient pipeline leak localization and automated isolation valve controller.

Real-time multi-parameter industrial effluent & drinking water quality buoy (Turbidity, TDS, pH, Dissolved Oxygen, Heavy-Metal conductivity anomaly) with automatic discharge shutoff.

Hardware-accelerated Maximum Power Point Tracking (MPPT) solar charge controller (Perturb & Observe / Incremental Conductance) handling partial shading or dual-axis solar tracking.

- Automated e-waste / plastic-polymer optical & inductive sorting mechanism for recycling plants.

TIER A: JUNIOR DIVISION (1ST & 2ND YEAR UG)

Working sensor-to-actuator hardware prototype demonstrating real-time physical measurement, automated control action (valve/relay/tracker actuation), and live telemetry.

TIER B: SENIOR DIVISION (3RD & 4TH YEAR UG / PG)

Closed-loop mathematical optimization (e.g., MPPT efficiency %, leak distance estimation algorithm, or multi-sensor calibration curve), low-power solar/battery field operation mode, and stress-tested hardware execution.

SLIDE 5 **MANDATORY PROOF-OF-COMPETENCE ARTIFACT (FOR ROUND 1 PPT)**

Your Slide 5 MUST include: **(a)** Complete sensor conditioning & actuator driver schematic, **(b)** Calibration equations or control algorithm flowchart, and **(c)** Energy/power budget calculation for autonomous field deployment.

TRACK 05: OPEN SILICON & HARDWARE INNOVATION

VTX-T5-03 Open Innovation in Defense, Aerospace & Secure Communications

DOMAIN TRACK	**Track 05: Open Silicon & Hardware Innovation**
ORGANIZING HOST	**Department of EE - VDT, Sri Shakthi Institute of Engineering and Technology**
CATEGORY	**Defense Electronics, Tactical Comms, Anti-Drone Systems & Fault-Tolerant Hardware**

1. BACKGROUND & INDUSTRY CONTEXT

REAL-WORLD CHALLENGE

Modern border security, disaster search-and-rescue, and aerospace systems operate in contested environments where GPS signals are jammed or spoofed, standard cellular/Wi-Fi links are unavailable, and rogue micro-drones pose perimeter security threats.

2. PROBLEM DESCRIPTION & SCOPE OF INNOVATION

TECHNICAL OBJECTIVE

Teams are invited to architect and prototype an **original defense, aerospace, tactical communication, or high-reliability fault-tolerant hardware/silicon system**.

Representative Focus Areas (Teams may propose these or any original defense/aerospace concept):

Acoustic / RF / Optical micro-drone detection, bearing estimation, and perimeter alerting array.

GPS-Denied Dead-Reckoning Navigation Node fusing IMU (Accelerometer/Gyro/Magnetometer), barometer, and optical/wheel odometry with drift correction.

Frequency-hopping or AES-encrypted tactical LoRa/RF field communicator for disaster-struck or zero-network border zones.

- Triple-Modular Redundant (TMR) or radiation-hardened avionics flight-controller voter architecture in FPGA/RTL that survives single-module hardware failure without glitching.

TIER A: JUNIOR DIVISION (1ST & 2ND YEAR UG)

Functional hardware or RTL simulation prototype demonstrating the core tactical sensing, encrypted transmission, navigation calculation, or fault-voting mechanism.

TIER B: SENIOR DIVISION (3RD & 4TH YEAR UG / PG)

Resilience against active jamming/interference/fault injection, cryptographic or mathematical error-bound analysis, and live end-to-end hardware or FPGA demonstration.

SLIDE 5 **MANDATORY PROOF-OF-COMPETENCE ARTIFACT (FOR ROUND 1 PPT)**

Your Slide 5 MUST include: **(a)** System threat model or fault-tree analysis, **(b)** Complete hardware/RTL block diagram, and **(c)** Link-budget, drift-rate, or fault-recovery timing calculations.

TRACK 05: OPEN SILICON & HARDWARE INNOVATION

VTX-T5-04 Open Innovation in EDA Automation, Scripting & Verification Tooling

DOMAIN TRACK	**Track 05: Open Silicon & Hardware Innovation**
ORGANIZING HOST	**Department of EE - VDT, Sri Shakthi Institute of Engineering and Technology**
CATEGORY	**Electronic Design Automation (EDA), Verification Automation, Compilers & Developer Tools**

1. BACKGROUND & INDUSTRY CONTEXT

REAL-WORLD CHALLENGE

In commercial semiconductor companies, verification and physical design engineers spend 70% of their project timeline writing repetitive SystemVerilog testbenches, parsing million-line STA timing/DRC log files in Perl/TCL/Python, debugging `.vcd` waveforms manually, and fixing basic RTL lint bugs (such as inferred latches, combinational loops, and unsynchronized clock-domain crossings).

2. PROBLEM DESCRIPTION & SCOPE OF INNOVATION

TECHNICAL OBJECTIVE

Teams are invited to build a **high-impact Electronic Design Automation (EDA) Software Tool, RTL Linter, Automated Testbench/Assertion Generator, Waveform Analyzer, or Synthesis/Layout Scripting Framework** that accelerates the hardware design and verification workflow.

Representative Focus Areas (Teams may propose these or any original EDA tooling concept):

Automated Verilog/SystemVerilog Static Linter & CDC Checker that parses RTL ASTs (Abstract Syntax Trees) and visually flags inferred latches, undriven nets, blocking-in-sequential bugs, and multi-clock hazards.

Automated UVM / Self-Checking Testbench & SystemVerilog Assertion (SVA) Generator from a simple module port specification or timing diagram.

Intelligent `.VCD` Waveform Debugger & Protocol Decoder that automatically spots bus protocol violations (e.g., AXI/APB/I2C handshake errors) and highlights the exact failing clock cycle.

- Automated RTL-to-GDSII PPA Space Explorer (Python/TCL wrapper around Yosys/OpenROAD/Cadence/Synopsys logs) that sweeps clock constraints and plots Pareto-optimal Area vs. Power vs. Delay curves.

TIER A: JUNIOR DIVISION (1ST & 2ND YEAR UG)

Working parser/automation tool with a clean CLI or GUI that ingests real `.v` / `.sv` / `.vcd` / `.log` files and outputs accurate bug reports, generated testbenches, or visual analytics.

TIER B: SENIOR DIVISION (3RD & 4TH YEAR UG / PG)

Full AST-based parsing (e.g., using PyVerilog / Verilator / Tree-sitter / custom parser rather than fragile regex), automated bug-fix patching or waveform violation pinpointing, verified against a suite of 10+ tricky buggy RTL test cases.

SLIDE 5 **MANDATORY PROOF-OF-COMPETENCE ARTIFACT (FOR ROUND 1 PPT)**

Your Slide 5 MUST include: **(a)** Your compiler/parser AST dataflow pipeline diagram, and **(b)** A concrete "Before vs. After" input RTL/Log/VCD snippet showing the exact output produced by your tool.

TRACK 05: OPEN SILICON & HARDWARE INNOVATION

VTX-T5-05 Wildcard Disruptive Silicon & Cyber-Physical Systems

DOMAIN TRACK	**Track 05: Open Silicon & Hardware Innovation**
ORGANIZING HOST	**Department of EE - VDT, Sri Shakthi Institute of Engineering and Technology**
CATEGORY	**Wildcard Open Innovation (Analog/Mixed-Signal, Neuromorphic, Quantum, Robotics & SoC)**

1. BACKGROUND & INDUSTRY CONTEXT

REAL-WORLD CHALLENGE

Breakthrough engineering often crosses traditional domain boundaries—combining Neuromorphic Spiking Neural Networks (SNNs), Analog/Mixed-Signal IC design, Quantum circuit emulation, Swarm Robotics, or novel hardware-software co-design into architectures that do not fit into standard predefined boxes.

2. PROBLEM DESCRIPTION & SCOPE OF INNOVATION

TECHNICAL OBJECTIVE

Have a bold, original hardware, VLSI, FPGA, embedded, or cyber-physical engineering project that falls outside `VTX-T1-01` through `VTX-T5-04`? `VTX-T5-05` **is our official Wildcard Disruptive Innovation Arena.** Teams may define their own problem statement provided it involves substantial hardware, silicon, RTL, embedded, or edge-systems engineering (pure CRUD websites or generic management apps without a hardware/silicon/edge-compute core are not eligible).

TIER A: JUNIOR DIVISION (1ST & 2ND YEAR UG)

To ensure total fairness alongside predefined problem statements, Wildcard teams are held to a strict self-definition standard:

1. **Quantified Problem Statement:** You must prove with real-world data why your chosen problem is an important engineering bottleneck.
2. **Measurable Technical Benchmarks:** You must define 3 concrete quantitative metrics (e.g., clock frequency, power in mW, latency in ms, SNR in dB, or accuracy %) by which the jury will judge your build at Hour 24.
3. **Live Twist Readiness:** Like all other tracks, shortlisted Wildcard teams will receive a custom **Day-1 Live Engineering Twist** tailored specifically to their submitted architecture during the December 15–16 offline finale at SIET.

TIER B: SENIOR DIVISION (3RD & 4TH YEAR UG / PG)

Evaluation strictly normalized relative to academic standing and engineering depth.

SLIDE 5 **MANDATORY PROOF-OF-COMPETENCE ARTIFACT (FOR ROUND 1 PPT)**

Your Slide 5 MUST include: **(a)** Complete architectural schematic, RTL block diagram, or circuit topology, **(b)** Preliminary simulation waveform, mathematical proof, or prototype benchmark proving your idea is physically buildable within 24 hours, and **(c)** Your 3 self-defined quantitative pass/fail target metrics.